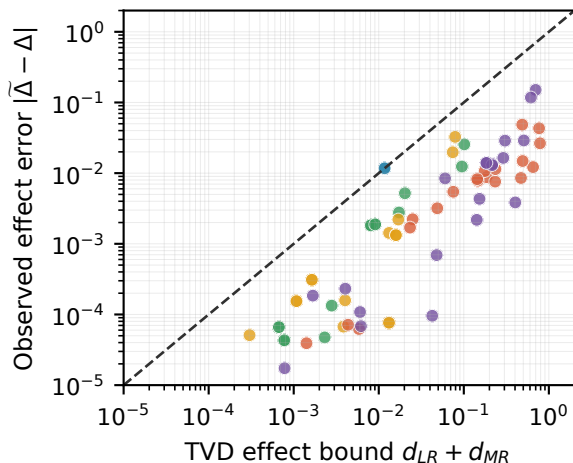
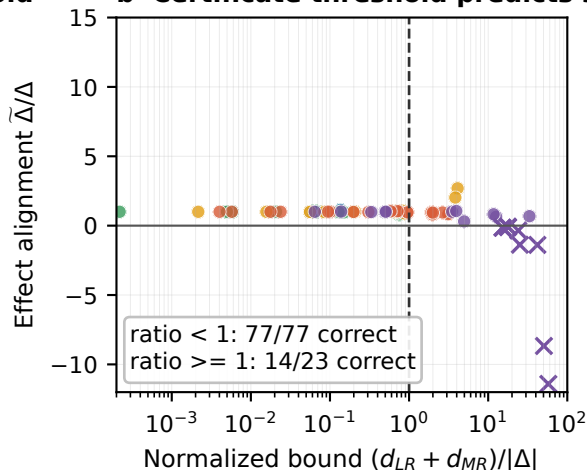


● karate (3q) ● chesapeake (7q) ● football (7q) ● ibm32 (18q) ● aves (24q) --- theorem boundary

a All 100 event-error bounds hold



b Certificate threshold predicts sign



c Per-case replication across backend and ordering

